

The dwarf planets

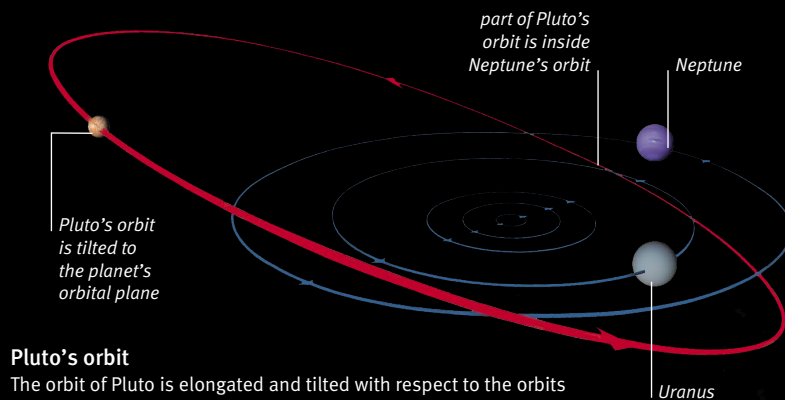
Eris, Pluto, and Ceres are the three dwarf planets in the solar system. They have existed for 4.6 billion years but have only been categorized as dwarf planets since 2006. The distant, icy worlds of Eris and Pluto orbit beyond Neptune. Ceres is much closer to the Sun in the main belt of asteroids.

DWARF WORLDS

Eris is the largest dwarf planet, and it is believed to be a mix of rock and ice with an icy surface. Its size is uncertain, with estimates ranging from 1,500 miles (2,400 km) to 1,850 miles (3,000 km) across. Eris was discovered in 2005 when images taken in 2003 were reanalyzed. At the time, it was almost 10 billion miles (16 billion km) away and the most distant solar system object seen. It completes one circuit of its elongated and highly inclined orbit every 560 years.

Pluto is a cold world. A thin, icy crust covers a body made of probably 70 percent rock and 30 percent water-ice. At an average distance from the Sun of 3.7 billion miles (5.9 billion km), it is unsurprising that its surface temperature is about -382°F (-230°C). This 1,432-mile- (2,304-km-) wide body has three moons. The largest, Charon, is about half the size of Pluto itself. Each of the pair spins every 6.38 days, and because Charon orbits Pluto in the same time, the two bodies keep the same face toward each other. From its discovery in 1930 until August 2006, Pluto had planetary status. It was then the most distant and smallest of the planets.

Ceres has been known since 1801. It was the first asteroid discovered and remains the largest known; it measures 596 miles (960 km) across. Ceres is a rocky body with water-ice near the surface. It orbits the Sun every 4.6 years.

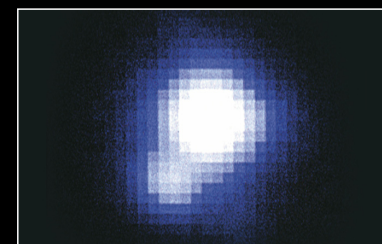


Pluto's orbit

The orbit of Pluto is elongated and tilted with respect to the orbits of the eight planets. Pluto completes one circuit in about 248 years, and for about 20 years of each orbit its path takes it closer to the Sun than Neptune. This last happened between 1979 and 1999.

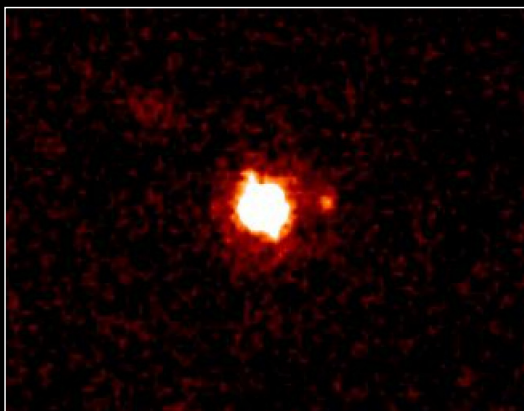
PLANETS OR DWARF PLANETS?

After the first Kuiper belt object was discovered, in 1992, Pluto's planetary status was questioned. The 2005 identification of an object larger than Pluto led to the creation of the "dwarf planet" class. This was established in August 2006 by the International Astronomical Union, the largest professional body for astronomers. Like planets, dwarf planets are almost round, but unlike planets, they have not cleared their neighborhoods: Ceres is in the main belt; Eris and Pluto are in the Kuiper belt. So far, only these three examples are known, but other likely candidates exist in the Kuiper Belt, and the number will rise.



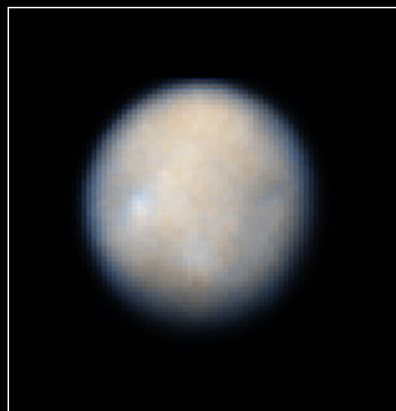
Ground view

This is how Pluto appears through the best ground-based telescopes. The bump to the lower left is its major moon, Charon.



Eris

This image from the 3-ft (10-m) Keck telescope in Hawaii has Eris at its center as a large bright object. The smaller, duller object to its right is Dysnomia. Keck astronomers realized the small dot was a moon when it moved with Eris against the stars.



Ceres

The Hubble Space Telescope studied Ceres for nine hours, the time it takes to make one orbit. It was found to be almost round: the diameter at its equator is wider than at its poles.

Pluto and its moons

On average, Pluto is a little under 40 times as far from the Sun as Earth. This image from the Hubble Space Telescope shows Pluto with Charon at its lower right and, farther right, the moons Nix (top) and Hydra (bottom).